

Syed Abbas Bukhari, PhD

Staff Scientist, Oncogenomics Laboratory — NCI / NIH

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PROFILE

Computational and systems biologist with 10+ years of experience integrating bulk, single-cell, and spatial transcriptomics with gene-regulatory network (GRN) modeling to identify lineage programs, prognostic cell states, and actionable dependencies in cancer. Current focus: pediatric solid tumors (rhabdomyosarcoma and pan-pediatric GRNs) at NCI; interested in extending these approaches to adult cancers and predictive modeling.

CORE EXPERTISE

- Tumor microenvironment & cell-state inference — single-cell atlas-based deconvolution of bulk RNA-seq; linking cell-state proportions to patient outcomes
- Gene-regulatory networks — ARACNe-AP / GENIE3 consensus GRNs, regulon activity, master-regulator and TF-target enrichment
- Multi-omics integration — bulk/scRNA-seq, spatial, CITE-seq; DepMap dependencies; drug resources (PRISM, ChEMBL, DrugBank, MIPE)
- Predictive analyses — feature selection, supervised models, and network-based prioritization for biomarker and target discovery
- Languages & tools — R, Python, Bash; Seurat, Scanpy, limma/DESeq2, GSEA, WGCNA; HPC/Slurm pipelines and AI based coding tools.

PROFESSIONAL EXPERIENCE

Staff Scientist — Oncogenomics Laboratory, NCI / NIH

May 2024 – Present

- Built a pan-pediatric consensus GRN (ARACNe-AP + GENIE3) from bulk RNA-seq across pediatric solid tumors; quantified TF/regulon activity via DEG enrichment to nominate lineage-specific master regulators.
- Integrated DepMap dependencies with PRISM, ChEMBL, DrugBank, and MIPE to prioritize druggable targets across tumor types (manuscript in preparation, co-first author).
- Deconvoluted bulk RNA-seq from rhabdomyosarcoma cohorts using a single-cell RMS atlas; identified prognostic cell states associated with patient's outcome.

Staff Scientist — Laboratory of Molecular Biology and Immunology, NIA / NIH

Feb 2023 – Mar 2024

- Led transcriptomic and regulatory-network analyses of immune aging across human and model-system datasets.

Postdoctoral Research Fellow — Carl R. Woese Institute for Genomic Biology, UIUC

Feb 2020 – Feb 2023 (concurrent remote appointment, York University, 2021–2023)

- Developed cross-species meta-analyses and behavior-related GRNs; co-authored work in Nature Ecology & Evolution and PNAS.

Postdoctoral Research Associate — Department of Integrative Biology, UIUC

Aug 2018 – Feb 2020

EDUCATION

- PhD, Informatics — University of Illinois Urbana-Champaign (2018). Advisor: Alison M. Bell.
- MS, Bioinformatics — University of Illinois Urbana-Champaign (2012).
- BS, Bioinformatics — COMSATS Institute of Information Technology (2009).

SELECTED PUBLICATIONS (FOR FULL LIST: SEE GOOGLE SCHOLAR)

- Traniello IM, Bukhari SA, Dibaenina P, et al. Single-cell dissection of aggression in honeybee colonies. *Nature Ecology & Evolution*. 2023.
- Sinha S, Jones BM, Traniello IM, Bukhari SA, et al. Behavior-related gene regulatory networks: a new level of organization in the brain. *PNAS*. 2020;117(38):23270–23279.
- Traniello IM, Bukhari SA, et al. Meta-analysis of honey bee neurogenomic response links Deformed wing virus type A to precocious behavioral maturation. *Scientific Reports*. 2020.
- Bukhari SA, Saul MC, James N, et al. Neurogenomic insights into paternal care and its relation to territorial aggression. *Nature Communications*. 2019;10(1).
- Saul MC, Blatti C, Yang W, Bukhari SA, et al. Cross-species systems analysis of evolutionary toolkits of neurogenomic response to social challenge. *Genes, Brain and Behavior*. 2018.
- Bukhari SA, Saul MC, Seward CH, et al. Temporal dynamics of neurogenomic plasticity in response to social interactions in male threespined sticklebacks. *PLOS Genetics*. 2017;13:e1006840.
- Rittschof CC, Bukhari SA, et al. Neuromolecular responses to social challenge: common mechanisms across mouse, stickleback fish, and honey bee. *PNAS*. 2014;111(50).
- Bukhari SA, Caetano-Anollés G. Origin and evolution of protein fold designs inferred from phylogenomic analysis of CATH domain structures in proteomes. *PLOS Computational Biology*. 2013.
- In preparation: A pan-pediatric gene-regulatory network reveals druggable dependencies across solid tumors (co-first author).

SELECTED HONORS

- HTAN Data Jamboree travel award (2025)
- Dissertation Completion Fellowship, UIUC (2017–2018)
- Travel Awards: Big Data Neuroscience Workshop (2017); Stickleback Behavior & Evolution Conference (2015); Computational Genomics Workshop, Broad Institute/MIT (2015)

SELECTED INVITED TALKS

- Allen Institute, Seattle — Neurogenomics of social behavior in sticklebacks (2019)
- University of Missouri, EEB Seminar (2019)
- University of Texas at Austin (2018)